

Enhanced Private Pilot Certification Course**LESSON 133 (1.0 GROUND, 1.0 DUAL NIGHT, and 5 LANDINGS)****Lesson Objective**

The purpose of this lesson is to introduce the student to the operational aspects of night flight. Emphasis will be placed on the additional planning necessary when operating at night. The flight will consist of both traffic pattern and practice area operations. A minimum of 10 takeoffs and landings to a FULL STOP must be completed by lesson 135.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Previous lesson was a solo flight.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 11 - Night Operations.
 - Chapter 18 - Emergency Procedures.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 17 - Aeromedical Factors

Lesson Content (Ground)**Aeromedical Factors (PHAK - Chapter 17)**

- Hypoxia - lack of oxygen
 - Hypoxic Hypoxia - insufficient oxygen available to the body.
 - Can be caused by a blocked airway.
 - Not the same amount of oxygen at higher altitude – same ratio, but just less air molecules.
 - Hypemic Hypoxia - blood is unable to transport oxygen.
 - Inability to absorb oxygen.
 - Caused by CO poisoning, blood loss, anemia.
 - Stagnant Hypoxia - blood is not flowing to move oxygen.
 - Inability to transport the oxygen.
 - Caused by a limb falling asleep.
 - Histotoxic Hypoxia - inability of cells to use oxygen.
 - Caused by usage of drugs and alcohol.
 - Symptoms of hypoxia (these will differ from person to person)
 - Cyanosis (blue nails and lips).
 - Headache.
 - Impaired judgment.

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- Takes longer to react.
 - Euphoria.
 - Visual impairment.
 - Drowsiness.
 - Lightheadedness or dizziness.
 - Tingling in fingers or toes.
 - Numbness.
- Corrective actions:
 - Use supplemental oxygen (requirements in [91.211](#)).
 - Decrease altitude.
 - Use medication.
 - Don't drink and fly.
 - Stretching – G suits.
- Hyperventilation – excessive rate and depth of respiration that leads to the abnormal loss of carbon dioxide from the blood or too much oxygen. Caused by anxiety, fear, or stressful situations.
 - Symptoms (can be similar to hypoxia):
 - Increased breathing rate.
 - Unconsciousness.
 - Visual impairment.
 - Lightheadedness or dizziness.
 - Tingling feelings.
 - Hot/cold sensations.
 - Muscle spasms.
 - Corrective actions:
 - Breathing normally.
 - Breath into paper bag.
 - Talking aloud.
- Middle ear and sinus problems – gasses trapped within the body expand with altitude causing pain in the middle ear and sinuses if gasses are not released. Usually caused due to change in pressure from climbing and descending with blockage.
 - Causes:
 - Illness and blockages.
 - Ear:
 - Pressure builds inside the middle ear – needs to be relieved when at altitude or when back on the ground.
 - Symptoms:
 - Can have loss of hearing.
 - Eardrum could rupture.
 - Infections.
 - Corrective actions:

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- Valsalva maneuver.
- Yawning.
- Swallowing.
- Sinuses (most frequently experienced during descent):
 - Symptoms:
 - Pain in sinuses.
 - Toothache.
 - Bloody mucus discharge.
 - Corrective actions:
 - Slow descent.
 - Don't fly.
- The Eustachian tube connects your throat with your ear – can cause you to be sick.
- Spatial disorientation - lack of orientation in regard to position, altitude, or movement of the airplane (confusion of sensory organs).
 - 3 systems:
 - Vestibular – organs in the inner ear that sense position by the way they are balanced.
 - 3 semicircular canals (based on 3 axes of motion) with fluid and tiny hairs that sense which way fluid is moving in the ear:
 - Horizontal.
 - Superior.
 - Posterior.
 - Illusions:
 - The leans – Once in a constant bank, (usually for 30 seconds), the vestibular system will “catch up” with the plane and the pilot will sense that the aircraft is straight and level.
 - Coriolis illusion – after banking for a while and making a sudden head movement, it can give the feeling of a maneuver or turn not occurring. Pilots might make wrong corrections and make things worse.
 - Graveyard spiral – prolonged coordinated turn can feel straight and level, could be due to the leans, (in a turn speed decreases and vertical lift decreases). The pilot will sense the altitude loss and add elevator pressure. But because the aircraft is actually banked, the turn will steepen, and the nose will drop more. Leads to an uncontrolled spiral dive.
 - Graveyard spin – enter a spin which is maintained and felt to decrease, so the opposite rudder is applied and then it feels like a spin in the opposite direction which will be corrected by use of rudder and then a spin in the original direction.
 - Somatosensory – nerves in the skin, muscles, and joints, which, along with hearing, sense position based on gravity, feeling and sound.

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- Illusions:
 - Somatogravic illusion – rapid acceleration gives the feeling of being nose-up. Pilots may push the nose down unnecessarily.
 - Inversion illusion – abrupt change from climb to straight and level can give the feeling of tumbling backwards. Pilot may push the aircraft nose lower.
 - Elevator illusion – abrupt vertical acceleration may give false feeling of being in climb causing pilot to push nose lower.
- Visual – eyes, which sense position based on what is seen.
 - Illusions:
 - False horizon – clouds or other obscured horizons can give the illusion of a false horizon. This can lead to the pilot using the wrong horizon reference line.
 - Runway width - A narrower-than-usual runway can create an illusion that the aircraft is at a higher altitude than it actually is. This can lead to the pilot flying a lower approach or rounding out too late. A wider runway can have the opposite effect with the risk of the pilot leveling out the aircraft high and landing hard or overshooting the runway.
 - Runway slope - an upsloping runway can create an illusion that the airplane is higher than it actually is. This can lead to the pilot flying a lower approach. A down sloping runway can create an illusion that the airplane is lower than it actually is. This can lead to the pilot flying a higher than normal approach.
 - Featureless terrain - An absence of ground features (like over the water, dark areas, desert, etc.) can create an illusion that the airplane is higher than it actually is. This can lead to the pilot flying lower or a lower approach.
- What is motion sickness?
 - Caused by the brain receiving conflicting messages about the state of the body.
 - Symptoms:
 - Loss of appetite.
 - Dry mouth.
 - Sweating.
 - Nausea.
 - Headaches.
 - Corrective actions:
 - Open air vents.
 - Focus on objects outside of the airplane and in the distance.
 - Avoid unnecessary head movements.
- What is carbon monoxide poisoning?

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- Colorless, tasteless, odorless gas that prevents hemoglobin from carrying oxygen to the cells.
 - Causes
 - Inhalation of combustion fumes.
 - Cars, planes, charcoal grills.
 - Symptoms
 - Headache.
 - Drowsiness.
 - Dizziness.
 - Corrective actions
 - Shut off heater.
 - Open air vents.
 - Seek medical attention.
- Fatigue and Stress
 - Fatigue
 - Acute – felt after long periods of physical strain.
 - Short term.
 - Coordination and alertness may be reduced.
 - Prevented by adequate sleep, exercise, nutrition.
 - Chronic fatigue - occurs by not recovering from acute fatigue.
 - Long term.
 - Performance can decrease.
 - Prolonged periods of rest are needed.
 - Stress:
 - Body's response to demands placed on it.
 - Physical and psychological.
 - Small levels are ok.
 - Can be a burst of energy/motivation.
 - Avoid by exercising, well nourished, and well rested.
- Dehydration and heatstroke – lack of fluids for the body to carry for normal functions.
 - Dehydration – occurs by inadequate intake, or loss of, fluids.
 - Drink plenty of fluids.
 - Avoid caffeine and alcohol intake.
 - Symptoms:
 - Headaches.
 - Nausea.
 - Internal organs can shut down.
 - Can lead to heatstroke .
 - Heatstroke – body is unable to keep itself cool.
 - Similar symptoms to dehydration but can lead to complete collapse.

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- Effects of alcohol and drugs – FAR [91.17](#):
 - Never combine drugs, alcohol, and flying.
 - 1 oz. of liquor, 1 bottle of beer, or 4oz. of wine can impair flying abilities.
 - Maximum .04 BAC.
 - Can increase the effects of hypoxia and dehydration.
 - FAA prohibits cockpit operations if you consume alcohol within 8 hours (12 hours for FIT).
 - If medication must be taken, ensure it is approved by the FAA (ask AME).
- Effects of nitrogen excess during scuba dives
 - Nitrogen is absorbed into the bloodstream during dives due to increased pressure and can expand and cause severe pain.
 - After uncontrolled ascent:
 - Wait at least 12 hours to fly up to 8,000 ft. after diving.
 - Wait at least 24 hours before flights above 8,000 ft.
 - After controlled ascent:
 - Wait 24 hours before flying at any altitude.
- Types of decompression sickness – nitrogen bubbles.
 - Bends – joints, elbows, shoulders, lips.
 - Chokes – in the lungs.
 - Skin – itching on the skin, swelling and skin compressions.
 - Neurologic – Brain, spinal cord, and peripheral nerves.
 - Recovery techniques:
 - Land as soon as possible.
 - Declare emergency if needed.
 - Barometric chamber if needed.

Night Operations

- Definitions of nighttime
 - FAA definition (FAR Part 1)
 - for logging nighttime
 - end of evening civil twilight to the beginning of morning civil twilight.
 - Civil twilight is defined in the Airman's Almanac.
 - A much more readable and useful source to find this information is airnav.com on the right-side bar below the airport diagram.
 - Position lights ([91.209](#))
 - must be on between sunset and sunrise.
 - Nighttime passenger currency (FAR Part 61 - 61.57): currency can be accomplished between 1 hour after sunset and 1 hour before sunrise. The minimum of 3 landings must be to a full stop.
 - Notes:
 - Nighttime hours can be found in the Airman's Almanac.
 - Alaska requires position lights "during the period a prominent unlighted object

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cannot be seen from a distance of 3 statute miles or the sun is more than 6 degrees below the horizon" (91.209).

- Night vision
 - How do the eyes work?
 - Rods and cones:
 - Two types of light nerve endings which transmit messages to the brain via the optic nerve.
 - Cones – Responsible for color, detail, and far away objects.
 - The cones are located in the center of the retina.
 - The fovea is where almost all of the light-sensing cells are cones.
 - This is the center of vision.
 - Need light in order to function.
 - Rods – Responsible for peripheral vision and vision in dim light.
 - The rods are located around the cones (peripherals).
 - Scan using off center viewing at night.
 - When attempting to find traffic do not stare directly at it, look slightly off to the left or right to allow the rods to see the aircraft.
 - The rods can take approximately 30 minutes to fully adapt to the dark.
 - Therefore, it is important to avoid bright lights before and during flight.
 - This is why red flashlights are recommended during flight, they do not disrupt the rod's dark adaptation as much.
 - Due to the increase in the body's need for oxygen at night (increased eye muscle usage, concentration, etc.), the AIM recommends using supplemental oxygen at or above 5,000ft.
 - AIM 8-1-2.
- Night illusions
 - Autokinesis
 - Caused by staring at a single point of light against a dark background creating the illusion that the light appears to move.
 - Prevent this by focusing the eyes on objects at varying distances and avoid fixating.
 - Keep the eyes moving and offset/use peripherals.
 - False horizon
 - Caused when the natural horizon is obscured/not readily apparent (night sky and ocean)
 - Generated by confusing bright stars and city lights.
 - Use and trust your instruments to maintain orientation.

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- Featureless terrain / black hole approach
 - An absence of ground features can create the illusion that the aircraft is higher than it actually is.
 - This results in a tendency to fly a lower than normal approach.
 - Use and trust your instruments to maintain orientation.
- Ground lighting
 - Regularly spaced lights along a road/highway etc. can appear to be runway lights.
 - Bright runway or approach lights can create the illusion the airplane is closer to the runway
 - Mitigate this as much as possible by maintaining situational awareness.
 - Know what to expect to see (type of airport/runway lighting), where you expect to see it and know where you are (use nav aids, GPS, landmarks, etc).
- Lighting
 - Aircraft lights:
 - 91.205 requires the airplane to be equipped with the following lights at night:
 - Landing light (electric) - if the operation is for hire.
 - Anti-collision lights.
 - Position lights.
 - Airport lighting (AIM 2-1):
 - Taxiway lighting:
 - Edge lights - blue.
 - Centerline lights - green, alternates green/yellow on the runway (lead off lights).
 - Runway guard lights - yellow.
 - Taxiway clearance bar lights - yellow.
 - Approach lighting:
 - VASI/PAPI: red and white.
 - Refer to chart supplement to see what approach lighting system and glidepath indicator an airport is equipped with.
 - Runway lighting:
 - Entry threshold - green.
 - Runway edge lights - white.
 - Last 2,000ft - amber.
 - Centerline lights - white.
 - Last 3,000ft - red and white.
 - Last 1,000ft - red.
 - Runway touchdown zone lights - white.
 - Runway end identifier lights - red.
- Preparation and required equipment:
 - Preflight inspection - 91.103:

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- Required equipment for VFR flight at night:
 - ATOMATOFLAMES and FLAPS – 91.205 Parts B and C.
 - Fuses (if applicable).
 - Needed – 1 set of spare fuses or 3 spare fuses for each kind required.
 - Landing Light (electric) - for hire.
 - Anti-collision light.
 - Position Lights.
 - Source of Power.
- Pilot equipment:
 - Flashlight: red and white
 - White light is used to preflight the airplane while the red light is used while in flight.
 - Again, red light will not impair night vision.
- Walk around
 - The preflight inspection is still necessary.
 - White light flashlight is good for the inspection (red light inside the cockpit).
 - Check all aircraft lights.
 - Check the ramp for obstructions.
 - Check NOTAMs, weather, obstructions, navigation systems and radios.
 - Fuel minimums at night are 45 minutes (FAA) and 60 minutes (FIT).
- In-flight orientation:
 - Relying on instruments at night is needed due to the absence of visual references.
 - Be familiar with runway/airport lights: know what you are looking for!
 - Use the HSI to bug in the runway heading.
 - Don't be afraid to ask ATC for help.
 - Pilot Controlled Lighting:
 - High Intensity Runway Lighting – 7 clicks.
 - Medium Intensity Runway Lighting – 5 clicks.
 - Low Intensity Runway Lighting – 3 clicks.
 - Round out and flare:
 - Judgment of height, speed, and sink rate may be impaired.
 - There is often a tendency to round out too high.
 - A good rule is to start the round out when the landing lights reflect on the tire marks on the runway.
- Visual scanning and collision avoidance
 - Emphasize that it is the pilot in command's responsibility to avoid traffic.
 - IF YOU SEE SOMETHING, SAY SOMETHING!
 - At night, the use of peripheral vision is the best method to scan for traffic.

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- Use the same scanning method but paying more attention to your peripheral vision. If a plane is spotted, avoid looking directly at it. Look slightly to the side of it.

Emergency equipment and survival gear

- Fire extinguisher:
 - Check in preflight to make sure it is properly charged (in the green), the pin is in, and it is latched.
 - Operation: open the latch, pull the pin out, point at the base of the fire and spray. If used in flight, make sure the vents and windows are opened.
- Survival kit:
 - A survival kit should provide sustenance, shelter, medical care and a means to summon help (radio, phone). Consider the terrain, the climate, and type of emergency communication equipment needed:
 - Winter: include blankets, extra clothes and jackets if available
 - Summer: blankets might be a good idea because it can still get cool at night
 - Mountainous terrain in December requires different survival gear than ocean flying in August.
- Overwater operations:
 - 91.509:
- Emergency locator transmitter (ELT):
 - These are designed to transmit a signal on 121.5 at a frequency of 243 megahertz (406 on newer ELTs).
 - 121.5: sounds like a loud beeping noise. The closer you are, the louder the sound.
 - 243/406 mhz: sends a signal containing the aircraft's location, pilot's name, contact information, and type of aircraft.
 - Usually armed to go off at an impact of around 5G's. Can also be manually turned on in the cockpit (ARM vs ON).
 - Batteries must be replaced after one hour of cumulative use or half the useful life is up.
 - Inspection is every 12 calendar months.

Go/no-go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.

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- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Flight)

Normal Takeoffs

- The takeoff procedure at night is the same as during the day.
- Areas of emphasis:
 - Clear the area thoroughly! It's harder to see at night!
 - Ensure the magnetic compass and HSI match with the intended runway centerline.
 - Verify line-up check is complete.
 - Perception of runway width, airplane speed, the flight attitude will vary at night.
 - The flight instrument should be checked frequently.
 - To ensure the airplane is climbing, check the airspeed indicator, the VSI and the altimeter; the darkness makes it hard to judge your rate of climb.
 - Turns should not be made until reaching a safe altitude.

Normal Landings

- A stable approach and landing should be made in the same manner as during the day.
- Areas of emphasis:
 - Clear the area thoroughly! It's harder to see at night!
 - Ensure the magnetic compass and HSI match with the intended runway centerline.
 - Use flight instruments more often (especially altimeter/airspeed indicator).
 - Distance, height, etc. may be deceptive.
 - Maintain specified airspeeds on each leg and watch the VSI to keep the approach under control.
 - If there are no centerline lights, align the airplane between the edge lights
 - Correct for any crosswind!
 - Use approach lights (VASI, PAPI, etc.) to maintain the desired glide path.
 - Judgment of height, speed, and sink rate may be impaired.
 - There is often a tendency to round out too high.
 - A good rule is to start the round out when the landing lights reflect on the tire marks on the runway.

Stalls

- Student says, student does.
 - Verify correct procedure.
- Areas of emphasis (nighttime):
 - It's difficult to maintain a visual reference, make use of the flight instruments:
 - Heading bug on the HSI.
 - Altimeter and VSI.
 - Turn coordinator.

Maneuvering During Slow Flight

- Student does, student says:
 - Verify correct procedure.
- Areas of emphasis (nighttime):
 - It's difficult to maintain a visual reference, make use of the flight instruments:
 - Attitude Indicator.
 - Heading bug on the HSI.
 - Altimeter and VSI.
 - Turn coordinator.

Simulated electrical failure

- Simulate an electrical failure by dimming the PFD.
- Ensure that the student can run through the correct checklist.
- Ensure that the student is reducing the electrical load.
- Question the student on their decision making.
- Is this a land "as soon as possible" or land "as soon as practical" emergency?

Postflight procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 134.
- Assign homework:
 - Review night operations as this topic will be quizzed on next lesson.
 - Review cross-country flight planning procedures.
 - Review emergencies.
 - Make sure flight publications (charts, chart supplement, etc.) are up to date.

Completion Standards

This lesson is complete when the student is able to:

1. Perform preflight actions at night.
2. Conduct maneuvers at night.
3. Maintain directional control at all times during takeoffs and landings.
4. Identify and correct for night illusions with instructor guidance.